Question # 7 of 10 (Start time: 04:09:14 PM, 20 January 2024) **Total Marks:** 1

Consider a Poisson distribution for the tossing of a biased coin. The mean for this distribution is μ. and the standard deviation for this distribution is given by

Select the correct option

Reload Math Equations

- ○ μ
- ○ $\frac{1}{\mu}$
- ○ $\sqrt{\mu}$
- ○ μ^2

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 04:01:14 PM, 20 January 2024)

Total Marks: 1

Mean and variance of the poisson distribution are

Select the correct option

Reload Math Equations

- not equal
- less than
- greater than
- equal



Click to Save Answer & Move to Next Question

sec(s)

STA642 - Probability Distributions (Quiz No 3) Quiz Start Time: 04:00 PM, 20 January 2024

Question # 1 of 10 (Start time: 04:00:39 PM, 20 January 2024) Total Marks: 1

Exponential distribution is a Bivariate distribution

Select the correct option

Reload Math Equations

- True
- False



Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 03:59:05 PM, 20 January 2024) **Total Marks:** 1

Find the value of Γ(5/2)?

Select the correct option

Reload Math Equations

- $\frac{7}{4}\pi^{\frac{1}{2}}$
- $\frac{1}{4}\pi^{\frac{1}{2}}$
- $\frac{5}{4}\pi^{\frac{1}{2}}$
- $\frac{3}{4}\pi^{\frac{1}{2}}$



Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 03:52:48 PM, 20 January 2024)

If we increase the value of $\alpha = 1 to 5$ with the $\lambda = 1/2$ then the shape of the distribution turns in to --------- distribution.

Select the correct option

Reload

- flatter than before
- Normal
- negatively skewed
- positively skewed

Click to Save Answer & Move to

3:51
vulms.vu.edu.pk/Quiz/

In a Poisson Distribution, if mean (m) = e, then P (x)

Select the correct option

○	$\frac{x!}{e^{(x-m)}}$
○	$\frac{e^{(m-x)}}{x!}$
○	$\frac{x!}{e^{(m-x)}}$
○	$\frac{e^{(x-m)}}{x!}$

3:51

vulms.vu.edu.pk/Quiz/

STA642 - Probability Distributions (Quiz No 3)

Question # 4 of 10 (Start time: 03:51:04 PM, 20 January 2024)

Gamma function is defined as $\Gamma(\alpha) = {}_0\int^{\infty} x^{\alpha-1} e^{-x} dx$?

Select the correct option

○	True
○	False

3:50

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Question # 3 of 10 (Start time: 03:50:18 PM, 20 January 2024)

A random variable X has an exponential distribution with probability distribution function is given by
$f(x) = 3e^{-3x}$ for $x>0$

$= 0$ otherwise
Find probability that X is not less than 2?

Select the correct option

Relo

- $e^{-6}-1$
- e^{-6}
- $e^{-6}-3$
- e^{-3}

Click to Save Answer & Move

3:49

vulms.vu.edu.pk/Quiz/

STA642 - Probability Distributions (Quiz No 3) Quiz Start Time: 0

Question # 1 of 10 (Start time: 03:48:55 PM, 20 January 2024)

If

$$f_X(x) = \begin{cases} \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

then on the value of $\alpha = 1$, which distribution we obtained?

Select the correct option

- Normal
- Exponential
- Poisson
- None

Click to Save Answer &

1:04 DS 4G 29%

Quiz
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BC200411334: FIZA BATOOL

Time Left 89 sec(s)

STA642 - Probability Distributions (Quiz No 3)

Quiz Start Time: 01:04 PM, 20 January 2024

Question # 1 of 10 (Start time: 01:04:05 PM, 20 January 2024)

Total Marks: 1

Poisson distribution is applied for _____

Select the correct option

Reload Math Equations

- Uncertain Random Variable
- Irregular Random Variable
- Discrete Random Variable
- Continuous Random Variable

Click to Save Answer & Move to Next Question

1:05 DS 4G 29%

Quiz
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BC200411334: FIZA BATOOL

Time Left 86 sec(s)

STA642 - Probability Distributions (Quiz No 3)

Quiz Start Time: 01:04 PM, 20 January 2024

Question # 2 of 10 (Start time: 01:05:30 PM, 20 January 2024)

Total Marks: 1

Job arrive at a facility at an average rate of 5 in an 8 hour shift. The arrival of the jobs follows poisson distribution. The average service time of the job on the facility is 40 minutes. The service time follows an exponential distribution. Idle time(in hours) at a facility per shift will be

Select the correct option

Reload Math Equations

- $\frac{14}{3}$
- $\frac{7}{5}$
- $\frac{5}{7}$
- $\frac{10}{3}$

Click to Save Answer & Move to Next Question

1:07 DS 4G 28%

Quiz
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BC200411334: FIZA BATOOL

Time Left 89 sec(s)

STA642 - Probability Distributions (Quiz No 3)

Quiz Start Time: 01:04 PM, 20 January 2024

Question # 3 of 10 (Start time: 01:07:16 PM, 20 January 2024)

Total Marks: 1

A random variable X has an exponential distribution with probability distribution function is given by
$f(x)= 3e^{-3x}$ for $x>0$

= 0 otherwise
Find probability that X is not less than 2?

Select the correct option

Reload Math Equations

- $e^{-6}-1$
- e^{-3}
- e^{-6}
- $e^{-6}-3$

Click to Save Answer & Move to Next Question

1:07 DS 4G 28%
Quiz
vulms.vu.edu.pk

STA642 - Probability Distributions (Quiz No 3)

Question # 4 of 10 (Start time: 01:07:53 PM, 20 January 2024)

If 'm' is the mean of Poisson Distribution, the P(0) is given by _____

Select the correct option

○	m^{-e}
○	e^{m}
○	e^{-m} ✓
○	e

1:08 DS 4G 28%

Quiz
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BC200411334: FIZA BATOOL

Time Left 89 sec(s)

STA642 - Probability Distributions (Quiz No 3)

Quiz Start Time: 01:04 PM, 20 January 2024

Question # 5 of 10 (Start time: 01:08:53 PM, 20 January 2024)

Total Marks: 1

A mobile conversation follows an exponential distribution $f(x) = \frac{1}{3}e^{\frac{-x}{3}}$. What is the probability that the conversation takes more than 5 times?

Select the correct option

Reload Math Equations

- $e^{-5/3}$
- $e^{-1/3}$
- $e^{-3/5}$
- $5e^{-15}$

Click to Save Answer & Move to Next Question

1:09 DS 4G 28%

Quiz
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STA642 - Probability Distributions (Quiz No 3)

Question # 6 of 10 (Start time: 01:09:31 PM, 20 January 2024)

The mean of exponential distribution is given as _____

Select the correct option

○	$(\mu)^2$
○	μ
○	$\frac{1}{\mu^2}$
○	$\frac{1}{\mu}$



1:10 DS 4G 28%

Quiz
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BC200411334: FIZA BATOOL

STA642 - Probability Distributions (Quiz No 3)

Question # 7 of 10 (Start time: 01:10:13 PM, 20 January 2024)

The mean and variance of the Gamma distribution are

Select the correct option

- $E(x) = \frac{\alpha * \beta}{\lambda} and Var(x) = \frac{\alpha}{\lambda^2}$
- $E(x) = \frac{\alpha}{\lambda} and Var(x) = \frac{\alpha}{\lambda^2}$
- $E(x) = \frac{1}{\lambda} and Var(x) = \frac{\alpha}{\lambda^2}$
- $E(x) = \frac{\alpha}{\lambda} and Var(x) = \frac{1}{\lambda^2}$

1:11 DS 4G 27%

Quiz
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BC200411334: FIZA BATOOL

STA642 - Probability Distributions (Quiz No 3)

Question # 8 of 10 (Start time: 01:11:03 PM, 20 January 2024)

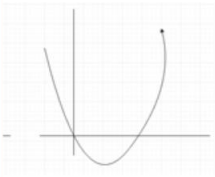
Which of the following graph represents gamma distribution?

Select the correct option

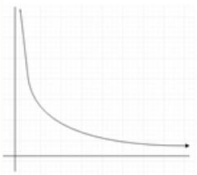
- Figure b



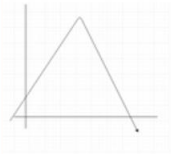
- Figure d



- Figure a



- Figure c



1:11 DS 4G 27%

Quiz
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BC200411334: FIZA BATOOL

STA642 - Probability Distributions (Quiz No 3)

Question # 9 of 10 (Start time: 01:11:55 PM, 20 January 2024)

The pdf of poisson distribution is

Select the correct option

- ○ $\frac{x!}{m^x e^{-m}}$
- ○ $\frac{e^{-m}m^x}{x!}$ ✓
- ○ $\frac{e^{-m}x!}{m^x}$
- ○ $\frac{e^{m}m^x}{x!}$

1:14 DS 4G 27%

Quiz
vulms.vu.edu.pk

STA642 - Probability Distributions (Quiz No 3) Quiz Start Time: 01:04 PM

Question # 10 of 10 (Start time: 01:14:06 PM, 20 January 2024)

If we increase the value of $\alpha = 1 to 5$ with the $\lambda = 1/2$ then the shape of the distribution turns in to --------- distribution.

Select the correct option

Reload

- negatively skewed
- Normal
- flatter than before
- positively skewed

Click to Save Answer & Move

STA642 - Probability Distributions (Quiz No 3)

Quiz Start Time: 06:27 PM, 19 January 2024

Question # 1 of 10 (Start time: 06:27:29 PM, 19 January 2024) **Total Marks:** 1

Sum of n independent Exponential random variables (λ) results in _____

Select the correct option

Reload Math Equations

- ○ Binomial random variable
- ○ Normal random variable
- ○ Gamma random variable
- ○ Uniform random variable

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 06:28:05 PM, 19 January 2024) Total Marks: 1

A random variable X has an exponential distribution with probability distribution function is given by
$f(x)= 3e^{-3x}$ for $x>0$

= 0 otherwise
Find probability that X is not less than 2?

Select the correct option

Reload Math Equations

- $e^{-6}-3$
- e^{-3}
- e^{-6}
- $e^{-6}-1$

Click to Save Answer & Move to Next Question

STA642 - Probability Distributions (Quiz No 3) **Quiz Start Time: 06:27 PM, 19 January 2024**

Question # 3 of 10 (Start time: 06:28:56 PM, 19 January 2024) **Total Marks:** 1

A mobile conversation follows an exponential distribution $f(x) = \frac{1}{3}e^{\frac{-x}{3}}$. What is the probability that the conversation takes more than 5 times?

Select the correct option

Reload Math Equations

- $e^{-3/5}$
- $5e^{-15}$
- $e^{-5/3}$
- $e^{-1/3}$



Click to Save Answer & Move to Next Question

STA642 - Probability Distributions (Quiz No 3)

Quiz Start Time: 06:27 PM, 19 January 2024

Question # 5 of 10 (Start time: 06:30:50 PM, 19 January 2024)

Total Marks: 1

The mean of exponential distribution is given as _____

Select the correct option

Reload Math Equations

- μ
- $\frac{1}{\mu^2}$
- $(\mu)^2$
- $\frac{1}{\mu}$



Click to Save Answer & Move to Next Question

STA642 - Probability Distributions (Quiz No 3)

Quiz Start Time: 06:27 PM, 19 January 2024

Question # 6 of 10 (Start time: 06:31:21 PM, 19 January 2024) **Total Marks:** 1

Consider a random variable with exponential distribution with $\lambda=1$. Compute the probability for $P(X>3)$?

Select the correct option

Reload Math Equations

- e^{-3}
- e^{-4}
- e^{-1}
- e^{-2}

Click to Save Answer & Move to Next Question

STA642 - Probability Distributions (Quiz No 3) **Quiz Start Time: 06:27 PM, 19 January 2024**

Question # 7 of 10 (Start time: 06:32:57 PM, 19 January 2024) **Total Marks:** 1

Exponential distribution is a Bivariate distribution

Select the correct option

Reload Math Equations

- ○ True
- ○ False



Question # 8 of 10 (Start time: 06:33:29 PM, 19 January 2024) **Total Marks:** 1

The mean and variance of the Gamma distribution are

Select the correct option

Reload Math Equations

- $E(x) = \frac{1}{\lambda} and Var(x) = \frac{\alpha}{\lambda^2}$
- $E(x) = \frac{\alpha * \beta}{\lambda} and Var(x) = \frac{\alpha}{\lambda^2}$
- $E(x) = \frac{\alpha}{\lambda} and Var(x) = \frac{1}{\lambda^2}$
- $E(x) = \frac{\alpha}{\lambda} and Var(x) = \frac{\alpha}{\lambda^2}$



Click to Save Answer & Move to Next Question

STA642 - Probability Distributions (Quiz No 3) **Quiz Start Time: 06:27 PM, 19 January 2024**

Question # 9 of 10 (Start time: 06:34:43 PM, 19 January 2024) **Total Marks:** 1

Poisson distribution is applied for _____

Select the correct option

Reload Math Equations

- ○ Uncertain Random Variable
- ○ Irregular Random Variable
- ○ Discrete Random Variable ✓
- ○ Continuous Random Variable

Click to Save Answer & Move to Next Question

sec(s)

STA642 - Probability Distributions (Quiz No 3) **Quiz Start Time: 06:27 PM, 19 January 2024**

Question # 10 of 10 (Start time: 06:35:16 PM, 19 January 2024) **Total Marks:** 1

Consider a Poisson distribution for the tossing of a biased coin. The mean for this distribution is μ. and the standard deviation for this distribution is given by

Select the correct option

Reload Math Equations

- μ^2
- μ
- $\frac{1}{\mu}$
- $\sqrt{\mu}$

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 06:31:21 PM, 19 January 2024) **Total Marks:** 1

Consider a random variable with exponential distribution with $\lambda=1$. Compute the probability for $P(X>3)$?

Select the correct option

Reload Math Equations

- e^{-3}
- e^{-4}
- e^{-1}
- e^{-2}

Click to Save Answer & Move to Next Question